Long memory.
Reliable computation.

Fractional diffusion in heterogeneous media

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Joint project with
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After the pump stops



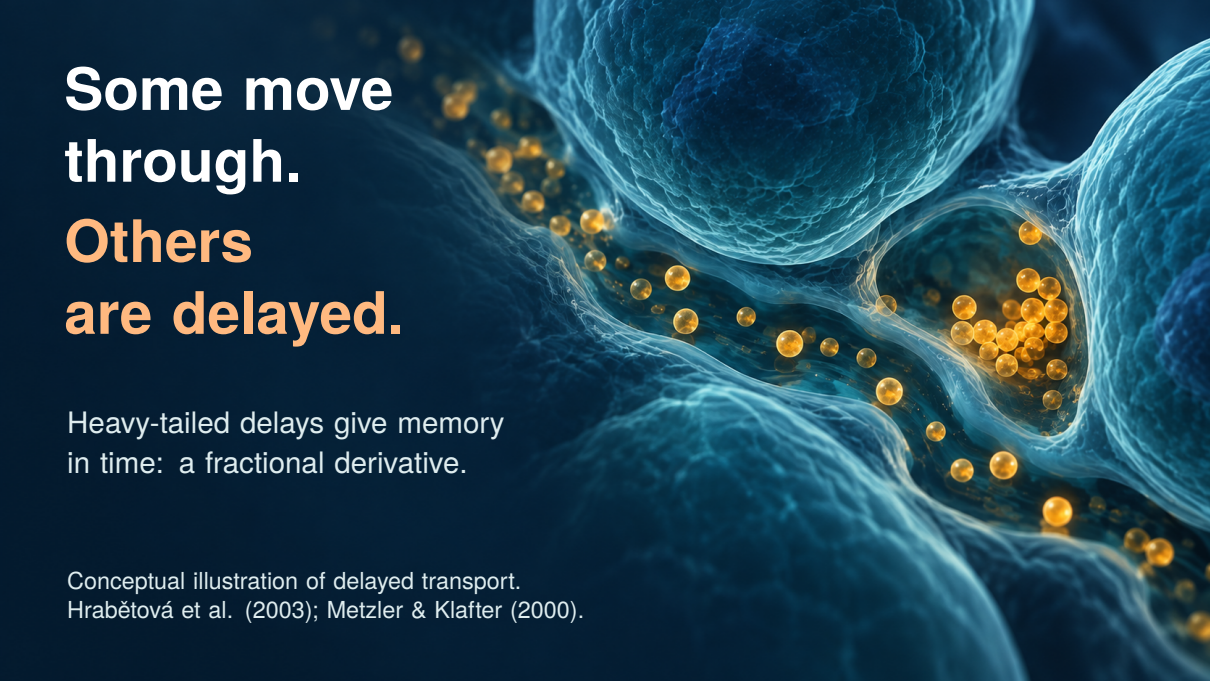
How much remains?
Where does it go?

Conceptual brain image.
Synthetic two-dimensional transport model.

Some move through.

Others are delayed.

Heavy-tailed delays give memory in time: a fractional derivative.



Conceptual illustration of delayed transport.
Hrabětová et al. (2003); Metzler & Klafter (2000).

Same dose. Different retention.

Retained dose at $T = 100$ · Infusion ends at $t = 1$

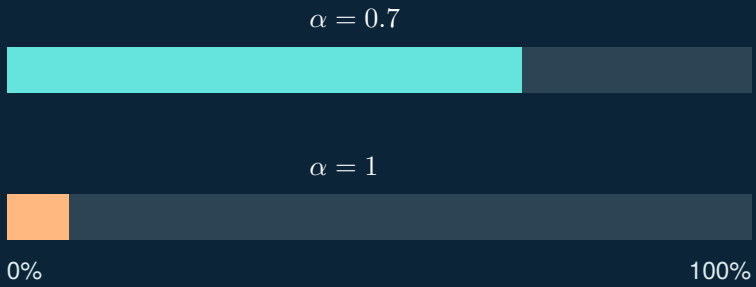
Synthetic 2D model · **Same physical pump**

Fractional memory

69%

Classical diffusion

8%



Is the difference real, or numerical error?

To answer it, a solver needs cheap memory and its own error check.

History becomes part of the state

Ordinary diffusion

Current rate

$$\dot{u} = Lu + f_1(t)$$

The current field and forcing determine the current change.

Fractional diffusion

Weighted history

$$D_*^\alpha u = Lu + f_\alpha(t)$$

Earlier changes also contribute. α controls their weighting.

Each step must account for the past.

Done directly, step n revisits all n earlier fields: N^2 work, N fields stored.

$0 < \alpha < 1$: memory acts in time. u: concentration; L: spatial diffusion; f_1, f_α: equation sources.

Where the error comes from

1 Space

Unresolved spatial structure
Refine the grid and compare.

2 Memory

Compressed kernel: several techniques
Check the fit and the recent history.

3 Time

Radau IIA with adaptive steps
Refine steps against a reference.

4 · SOLVE & RECOVERY

From residual to error

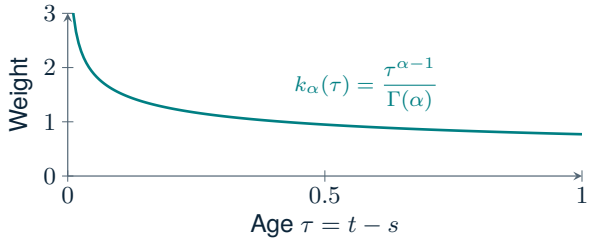
The solver controls a residual, but we need the concentration error.

endpoint error
$\leq h \times$ **residual**

Constant 1, sharp for any stiffness and any number of memory modes.

Memory is a weighted history

$$\underbrace{u(t)}_{\text{now}} = \underbrace{u_0}_{\text{start}} + \int_0^t \underbrace{k_\alpha(t-s)}_{\text{weight}} \underbrace{[Lu(s)+f(s)]}_{\text{past dynamics}} \, \mathrm{d}s$$



Recent past weighs most; older past still counts.

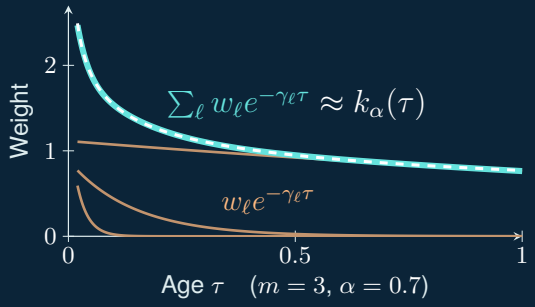
Shown: $\alpha = 0.7$.

A history at every spatial point.

u: concentration; L: spatial diffusion; f: input; $0 < \alpha < 1$. Integral form of the Caputo model.

Store m fields, not the whole history

Approximate the weight by a sum of simple decays, as in Guglielmi–Hairer (2025).



One decay, one memory field

$$\dot{z}_\ell = \underbrace{-\gamma_\ell z_\ell}_{\text{fade}} + \underbrace{Lu + f(t)}_{\text{new dynamics}}$$

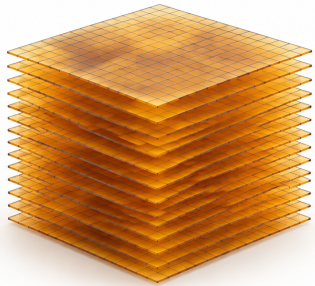
$$u = u_0 + \underbrace{\sum_\ell w_\ell z_\ell}_{\text{recovery}}$$

No history sums: each memory field obeys a local ODE.

Fit done for $\delta \le \tau \le T$; short-lag error counted separately. $w_\ell, \gamma_\ell > 0$, $z_\ell(0) = 0$.

Less history. Same spatial grid.

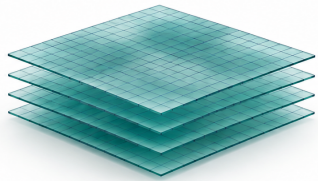
N_t **past fields**



$O(NN_t)$ storage

Full-history L1, 127^2: 4.2 GB, 30 min

m **current memory fields**



$O(mN)$ storage

Ours, 127^2: 93 MB, 22 s

Each field still contains N spatial values.

Both at about 10^{-7} error. Fast and slow memory rates call for a stiff integrator.

Only three fields are coupled per step

$3mN$ — memory-stage unknowns

exact elimination, any m $\longrightarrow$ within the compressed model

$3N$ — the 3 stage derivatives s

$$\mathcal{C}\boldsymbol{s} = \boldsymbol{d}, \qquad \mathcal{C} = I_3 \otimes I_N - hS_h \otimes L$$

$S_h = \sum_\ell w_\ell \, (I_3 + h\gamma_\ell \mathsf{A})^{-1}\mathsf{A}$: one 3×3 matrix for all modes.

Any linear L

Heterogeneous, nonseparable or nonsymmetric.

Fast preconditioner

Krylov on the full L, preconditioned by a sine transform: $O(N \log N)$.

Steady iterations

Count changes by at most one across grids and m.

A residual is not a concentration error

Residual: an equation imbalance. Error: a difference between solutions.

Reduced solve
Condensed residual

$$\boldsymbol{r} = \mathcal{C}\tilde{\boldsymbol{s}} - \boldsymbol{d}$$

The Krylov solver drives this down.

Memory recovery
Original residual

$$\boldsymbol{r}_\ell^{\mathrm{orig}} = \dot{\tilde{z}}_\ell + \gamma_\ell \tilde{z}_\ell - L\tilde{u} - f$$

Recovered stages put back into each memory equation.

Concentration
Physical-stage error

$$\widetilde{U} - U^*$$

What we need small.

Which residual predicts the error: $\boldsymbol{r}$ or $\boldsymbol{r}_\ell^{\mathrm{orig}}$?

Why the original residual can grow

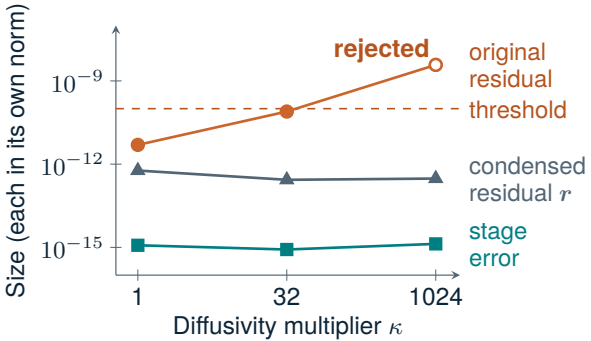
Exact recovery links the condensed and original residuals.

$$\underbrace{r_\ell^{\mathrm{orig}}}_{\text{original}} = h(\mathsf{A} \otimes L) \quad \underbrace{\boldsymbol{r}}_{\text{condensed}}$$

$$\|L\| \sim \kappa\, \Delta x^{-2}$$

κ multiplies the diffusivity: on this 7^2 grid, $\kappa = 1024$ is as stiff as 255^2.

At $\kappa = 32768$ the default test stops every run at startup.



Test steps, $\alpha = 0.7$, $t = 0.1$.

The default test rejects a step whose error is at roundoff.

Measured: $\boldsymbol{r}$ flat at $3\text{–}6 \times 10^{-13}$; the original residual grows $760\times$, past its 10^{-10} threshold.

The endpoint residual bound

$$\underbrace{\|U_3 - U_3^*\|_M}_{\text{error at the end of the step}} \quad \leq \quad 1 \cdot \underbrace{h\,\|\boldsymbol{r}\|_H}_{\text{step} \times \text{solver residual}}$$

Sharp constant 1
Even for arbitrarily stiff diffusion.

Uniform in the memory
Any number, weights and rates of modes.

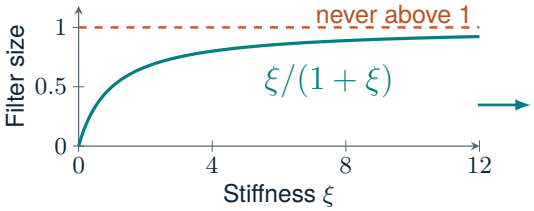
Computable
Known during the solve: a stopping target.

Answer: r, the solver's own residual, bounds the error.

Large residual, yet small error

$$U - U^* = h(\mathsf{A} \otimes I)\underbrace{(\mathcal{C}^{-1} - I)}_{\text{recovery filter}}\boldsymbol{r}$$

Scalar picture: $\mathcal{C} = 1 + \xi$



The full coupled system

$$\|\mathcal{C}^{-1} - I\|_H \leq 1$$

For any stiffness and any positive memory modes.

So the endpoint error is at most h times the residual.

One undamped, unit-weight mode: classical BS-stability (Frank et al. 1985; Dekker 1984).

Four recovery terms join the bound

The theorem assumes exact recovery; a computed step adds four defects.

$$\|\widehat{U} - U^*\|_H \le \underbrace{h\,a_\Pi \|\widehat{\boldsymbol{r}} + a_c\|_H}_{01} + \underbrace{\|\delta\|_M}_{02} + \underbrace{\textstyle\sum_\ell w_\ell \big(c_\ell \|e_\ell\|_H + \|\zeta_\ell\|_H\big)}_{03} + \underbrace{\|\eta\|_H}_{04}$$

01 Solve and assembly

$\widehat{\boldsymbol{r}}$: residual of the reduced solve;
a_c: stored-coefficient defect

02 Inherited field

δ: old field minus the field its memories give

03 Memory recovery

e_ℓ, ζ_ℓ: defects in mode ℓ's derivative and stage

04 Physical summation

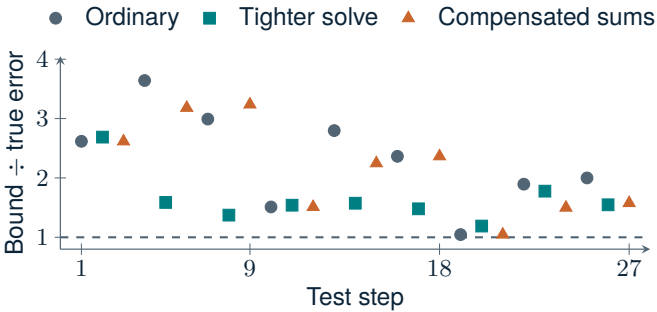
η: defect in the sum $U = u_0 + \sum_\ell w_\ell Z_\ell$

Every term can be evaluated so a complete error budget works.

$\widehat{U}$, U^*: computed and exact stages of the same step; H, M: stage-weighted and spatial mass norms; a_Π: a Radau constant; $c_\ell = \min\{h a_\Pi, \gamma_\ell^{-1}\}$.

The bound held on every test step

27 solver steps: the four-term bound vs the step's true error $\|\widehat{U} - U^*\|_H$.



1.046–3.640

bound ÷ true error, over all 27 steps

Always ≥ 1: the bound held.

On every step the bound held, and was at most 3.6× the true error.

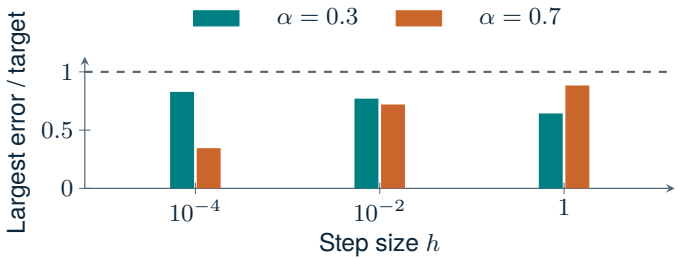
7^2 grid, $\alpha = 0.7$, $\kappa = 1, 32, 1024$; solve to 10^{-11}, to 10^{-13}, or 10^{-11} with compensated sums.

Stop at $h\|\boldsymbol{r}\| \le \tau$: error below τ

The bound turns the solver's residual into a stopping rule.

$$h\,\|\boldsymbol{r}_k\|_H \quad \le \quad \tau$$

τ: endpoint-error target. k: iteration index.



108

stopping decisions across 36 cases

0.883 τ

Largest observed endpoint error.

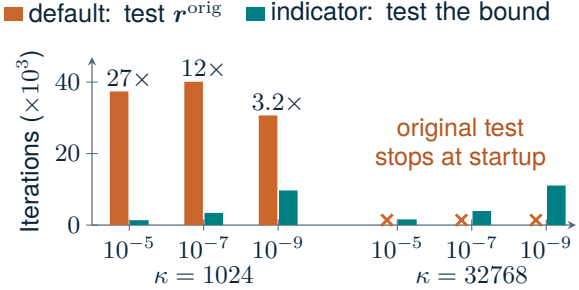
Never above the target, and within 12% of it: safe, not wasteful.

Three grids, two positive compressions, three step sizes and three tolerances.

Fewer rejections on stiff problems

Two ways to accept a step: test r^{orig} (default) or test the bound (indicator).

Stiff 7^2 family: iterations per tolerance



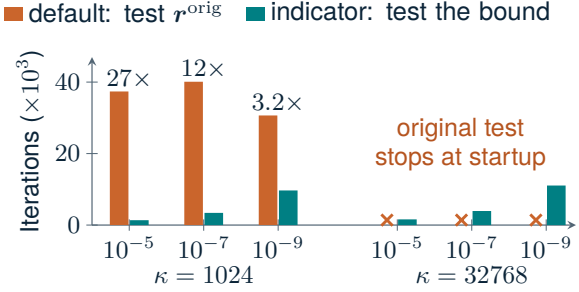
$\kappa = 1024$**: also 99% fewer rejections.**

Same solver, same accuracy: the indicator makes it 2–2.6× faster.

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Stiff 7^2 family: iterations per tolerance



$\kappa = 1024$: **also 99% fewer rejections.**

Infusion application

$\alpha = 0.7$, 511^2 grid, $T = 10$

47%

shorter run: 1.9 h → 1.0 h

758 to 0

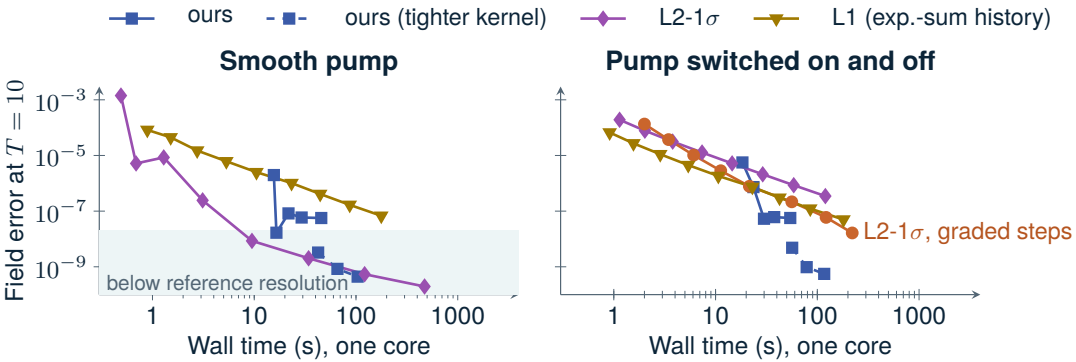
stage-test rejections

Exposure: $|\Delta E| = 4.7 \times 10^{-10}$.

Same solver, same accuracy: the indicator makes it 2–2.6× faster.

Fastest at high accuracy

Same problem, 127^2 grid and reference, one core. Middle: graded steps too.

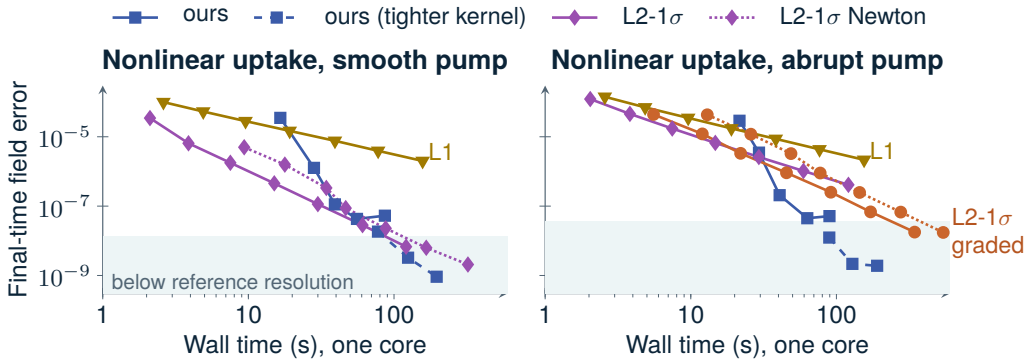


Smooth: second order is 2–4× faster at $T = 10$, level at $T = 100$.

Abrupt pump: ours is 4× faster at 10^{-7} and alone reaches 10^{-8}.

Fastest at high accuracy

Add saturable uptake $-Vu/(K+u)$. Same grid, one core.



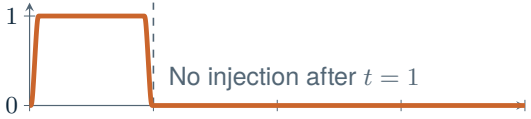
Nonlinear, smooth: linearized second order faster at 10^{-6}, level at $2 \cdot 10^{-8}$.

Nonlinear, abrupt: ours fastest from 10^{-6} on: $1.3\times$ at 10^{-6}, $2.9\times$ at 10^{-7}.

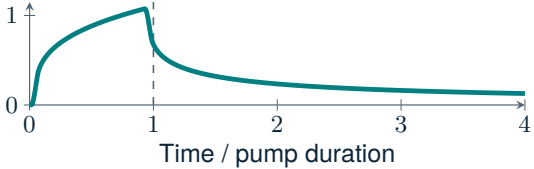
After shutoff, the source keeps a history

Same physical pump for all orders. Total injected dose $J = 11/12$.

Physical pump $Q(t)$



Caputo source $I^{0.3}Q(t)$, $\alpha = 0.7$



In the evolution equation

$$D_*^\alpha C = LC + \rho\, I^{1-\alpha} Q$$

ρ: injection profile (unit mass).

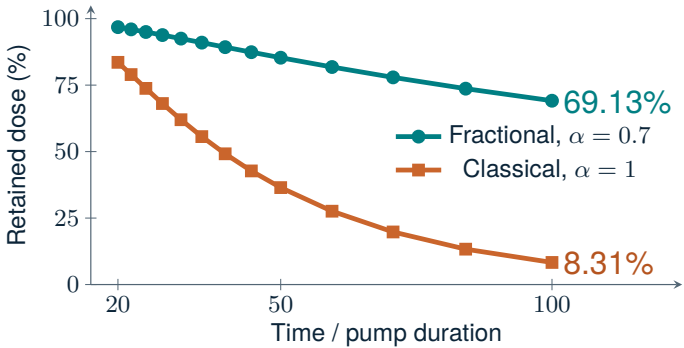
It remembers the injection;
no new material enters.

Regional exposure
Concentration integrated over time.

$$E_R(T) = \int_0^T \frac{1}{|R|} \int_R C(x,t)\,\mathrm{d}x\,\mathrm{d}t$$

The contrast is real, not numerical error

Retained dose after the pump stops at $t = 1$.



Central concentration

17.7×

the classical value at $T = 100$

The gap is $89\times$ its error budget.

The 69% vs 8% gap is $78\times$ the combined error budget.

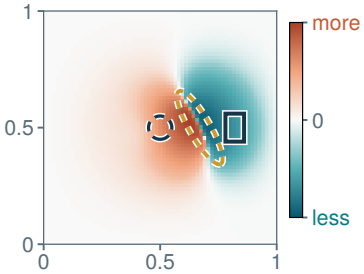
One 511^2 run to $T = 100$: 31 minutes, 477 steps, 158 memory modes.

A thin obstacle needs a fine grid

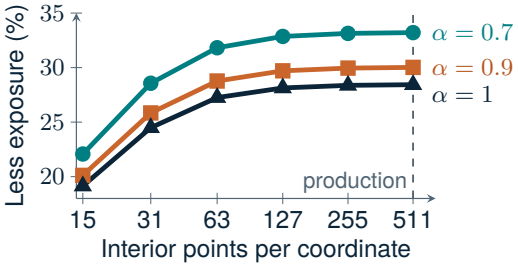
How much less material reaches the box behind a slow-diffusion obstacle?

With minus without obstacle

Circle: pump; ellipse: obstacle;
box: the region we measure.



Shielding of the box, against the grid



22% → 33% shielding at $\alpha = 0.7$, coarsest to finest grid.

A coarse grid misses a third of the shielding: we need 127^2 or finer.

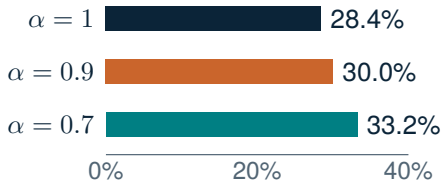
Shielding $= 1 - E_{\text{with}}/E_{\text{without}}$, E: box exposure at $T = 10$.

A small memory effect, resolved

Shielding of the box at $T = 10$, on 511^2 grids, for three orders.

Shielding behind the obstacle

Less exposure in the box than without the obstacle.



More memory, more shielding.

Our numerical error ($\pm 0.6\%$) is 4–11$\times$ smaller than the memory effect.

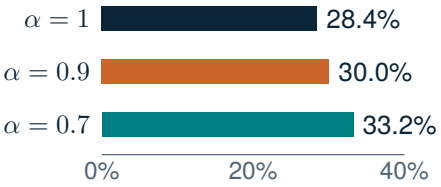
Extra shielding $= 1 - R_\alpha / R_1$, with $R = E_{\text{with}} / E_{\text{without}}$.

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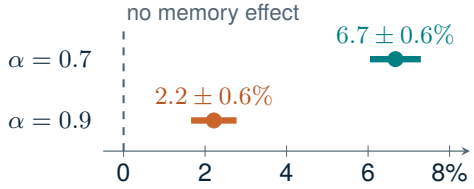
Less exposure in the box than without the obstacle.



More memory, more shielding.

Extra shielding from memory

Compared with classical diffusion ($\alpha = 1$), with the numerical uncertainty.



$\pm$: range allowed by both runs' error budgets.

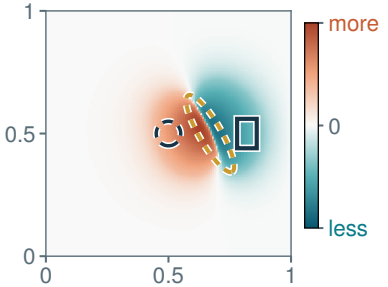
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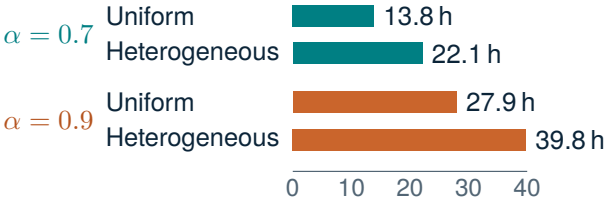
The same integrator runs in 3D

Slice $z = 0.5$, $T = 10$, $\alpha = 0.7$

Heterogeneous minus uniform.



Run time per case (hours), one core



Heterogeneous runs: 177 and 165 stage-test rejections (7 on 63^3), the kind the indicator removes in 2D.

154 memory modes: 2 million unknowns on one core, about 11 GB.

Model result on this grid: exposure ratio 0.74 at $\alpha = 0.7$ (2D: 0.67).

Summary of the talk

Memory costs three fields, not the history.
Exact elimination $3mN \to 3N$ for any m; 11 GB in 3D.
In 2D at 127^2: 93 MB of history instead of 4.2 GB.

The solver's residual bounds the error.
Error $\leq 1 \cdot h\|\boldsymbol{r}\|$, any stiffness, any number of modes.

Accept on the bound: no false rejections.
Versus the default test: 99% fewer stiff rejections;
4$\times$ faster than the best competitor at 10^{-7}, abrupt pump.

2 million
unknowns, one core

1
sharp constant

2–2.6$\times$
faster, equal accuracy

Next: an end-to-end bound for completely monotone memories and nonlinear reactions.
And run it on GPUs for real-life applications such as real brain anatomy.

Happy 60th, Nicola!

Thank you. Questions?

